

Intro to Bio-intensive Agriculture

In agriculture, each farm is different due to the unique set of conditions within which it operates. To become profitable, the use of equipment, space, and time available must be optimized.

The so-called "bio-intensive" model is first and foremost a **diversified organic cropping system that intensifies production with the goal of achieving maximum yields from a minimum area of land, while simultaneously increasing biodiversity and sustaining the fertility of the soil**. Although not all aspects discussed in this document are exclusive to this model, it seems logical that we discuss them here in order to clearly convey what we consider to be the foundations of our way of working.

Limited Mechanization

While large-scale farming models rely heavily on tractors and farm machinery, one of the most distinctive aspects of the bio-intensive model is its limited use of mechanization. Indeed, all operations on the bio-intensive farm (tillage, planting, weeding, harvesting) are done manually, using small tools or a walk-behind tractor.

This provides several advantages:

- The absence of tractors allows for more cultivation acreage as there is no need for roads or end-of-field turning areas.
- The absence of tractors limits the costs associated with their acquisition, maintenance and use.
- The absence of tractors limits the time spent in the garage for maintenance and repairs.
- The use of light equipment and tarps allows for early entry into the field.
- The use of light equipment limits soil compaction and its negative effects on crops.
- Intensification of spacing is not limited by mechanized equipment since planting, weeding and harvesting operations are all done manually.

Note that a tractor can still be useful for moving heavy loads or turning farm-made compost.

Soil Protection (Minimum Tillage and Continuous Cover)

In bio-intensive farming, no-till or minimum-till is preferred. This means avoiding turning over soil layers beyond 8 cm (3 in), and choosing other techniques such as the use of a broadfork and silage tarps. Minimal tillage maintains soil productivity by:

- Stabilizing organic matter in the soil.
- Reducing the weed seed bank.
- Promoting the health and proliferation of beneficial organisms.

In addition to minimizing tillage, we must make sure that the soil is covered at all times. Whether it is with a crop, mulch, green manure or a tarp, continuous soil cover is an iconic feature of the bio-intensive model as it promotes the establishment and maintenance of productive soil. Covering the soil:

- Limits the spread of weeds.
- Helps conserve water in the soil.
- Promotes biological activity (beneficial microorganisms, bacteria, fungi, insects) in the soil.
- Limits nutrient leaching.
- Limits wind and water erosion of the soil.

Standardized Systems

In the bio-intensive system, as many elements as possible are standardized in order to increase flexibility and optimize the efficiency of operations. This makes it possible to effectively plan the season and to reach objectives.

- Length and width of beds, width of aisles, number of beds per block.
- Dimensions of equipment (tarps, nets, row covers, irrigation lines).
- Technical itineraries, farming practices and procedures.

A standardized system is much easier to understand and control. Planning, communication and execution of daily, weekly and seasonal operations are also carried out more smoothly.

Permanent Beds

A permanent bed is defined as a production area that is in the exact same location year after year. As a result, the aisle will also always be in the same place. This can be achieved by using a system of markers or stakes that define the location of the beds.

This way of proceeding:

- Allows for improved production areas (the beds) each year.
- Allows for the efficient use of soil amendments (only the cultivated areas are fertilized, not the entire surface).
- Limits the compaction created by foot traffic to the aisles which are always in the same place.
- Facilitates the understanding and standardization of the system.

Hilling the beds in order to make raised beds (4-6 inches above the level of the aisles) is particularly useful and recommended in wet and cool conditions. It makes it possible to:

- Increase the depth of the top soil layer.
- Promote drainage.
- Accelerate the warming of the soil in spring.
- Efficiently mark out beds and walkways.
- Assist the accumulation of water on the tarps.

Hilling is done with a shovel or rotary plow, once a year at most, ideally after the last crop and before winter, thereby facilitating earlier cultivation in spring. This technique is not recommended for farms in hot, dry climates with adequately drained soils.

Intensive Spacing

Another distinctive aspect of the bio-intensive model is the high crop density. In a more extensive system, crops are weeded (and sometimes harvested) mechanically. It is these operations that limit crop density (one will rarely see more than three rows per bed in these systems). High planting density provides the following benefits:

- High yields per productive space used.
- Makes harvesting by hand more efficient.
- Faster in-row and inter-row soil coverage, resulting in reduced weed pressure.
- Increased efficiency of equipment (irrigation, tarps and nets).

Rapid Successions

A succession is defined as a sequence of crops planted in the same place, one after the other. In other words, in the bio-intensive system, the same bed will be cultivated several times in a row. For example, after opening the garden in the spring, radishes could be planted in a given bed. When all the radishes are harvested, the *same bed* would be prepared again and lettuce would be transplanted. When the lettuce is ready, all the heads are harvested, and we start again with arugula. Between each succession, a silage tarp is laid out on the beds so that they are ready for the next crop.

Depending on the geographical context, climatic conditions, available infrastructure (tunnels, greenhouses, etc) and the chosen crops, it is possible to produce between 1 and 4 crops per year on the same bed. This technique increases the productivity of each square meter of crop and decreases the total surface area required for the same production, which offers the opportunity to leave more space for biodiversity-friendly areas.

To increase the success and speed of successions, it is necessary:

- To know the dates of the last spring frost and the first fall frost.
- To choose crops with short days to maturity.
- To choose cultivars that produce constant and predictable results.

- To build a good garden plan so the use of tarps is optimized.
- To avoid letting the beds become overgrown with weeds.
- To harvest when the crop is mature and destroy the residues as soon as possible (mow if necessary and lay out silage tarp).

Greenhouses

The profitability of greenhouse crops has been proven. In the face of climate change, they ensure a continuous and high-quality production. Heated greenhouses allow for the production of high value crops with high yields and are an integral part of the bio-intensive farming landscape.

Conclusion

The implementation of the principles described here has the potential to help build a farming system that is highly profitable.

We hope that reading this document has given you a good understanding of the benefits of the bio-intensive model and will allow you to make informed choices for your farm business.